Aditya Kumar Pandey

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PROFILE SUMMARY

Machine Learning Engineer with 3+ years' experience building and deploying production ML and data systems across financial services and AI product development, from data pipelines (Python, SQL) to model deployment and MLOps (Docker, FastAPI, CI/CD, AWS/Azure/GCP). Currently an Analyst Intern at Delta Strategies, London, whilst completing an MSc in Data Science at LSE. Eligible for the UK Graduate Route visa (2-year, no sponsorship required).

EDUCATION

MSc Data Science — The London School of Economics and Political Science, London, UK *Sep 2025 – Sep 2026*
BTech Computer Science and Engineering — REVA University, Bengaluru, India *Aug 2018 – Jun 2022*

EXPERIENCE

Analyst Intern — Delta Strategies, London, UK *Jan 2026 – Present*

- The investment team relied on a manual, multi-day Excel reporting process for portfolio and risk analysis; owned end-to-end delivery of a production risk-management platform (Python, Streamlit) deployed on DigitalOcean, cutting reporting turnaround nearly 80%
- Margin, trading and hedge-fund data was scattered across disconnected sources; designed and built a centralised SQL data store to feed the platform with reliable, unified data
- Partnered directly with portfolio managers, traders and engineering to gather requirements and prioritise the platform roadmap, ensuring delivery matched the investment team's most pressing risk-reporting needs

Data Scientist — Capital Number, India *Feb 2025 – Aug 2025*

- Analysts spent significant time manually reviewing multi-modal (image, video, text) documents; built and deployed a RAG system on GCP using Gemini, with embeddings stored in Vertex AI Vector Search and orchestration via LangChain
- Engineered the embedding and indexing pipeline to handle unstructured multi-modal data at production scale, improving retrieval accuracy and cutting manual document processing by approximately 60%
- Partnered with product and engineering stakeholders to scope requirements and define success metrics, delivering the system to agreed timelines

Data Scientist — Wissen Research, India *Sep 2024 – Jan 2025*

- The research team spent significant manual effort extracting information from documents; automated workflows using LLM-driven information extraction, cutting manual effort by approximately 70% and shortening time to insight
- Built REST APIs in Flask to integrate the extraction models into internal tooling, achieving response times of approximately 50ms
- Collaborated with the research team to validate extraction outputs against domain requirements, ensuring reliability before workflow rollout

Data Scientist — Jupitice Justice Technology, India *Jan 2024 – Jun 2024*

- The legal team needed to scale client self-service support without adding headcount; engineered the retrieval layer of a client-facing RAG chatbot built on the OpenAI API
- Optimised retrieval using a TF-IDF model, improving efficiency by approximately 70%
- Iterated on retrieval logic with legal-team stakeholders to reduce irrelevant results and improve response quality

Software Developer — Cognida AI, India *Aug 2022 – Jan 2024*

- Logistics emails contained unstructured shipment data requiring manual review; fine-tuned a BERT model for named-entity recognition (NER), defining entity types with the client's ops team and labelling 1,000+ records to train it
- Built and containerised a FastAPI service backed by PostgreSQL to serve the model and store extracted entities and responses
- Deployed the service into the client's ops workflow, replacing manual email review with automated, structured entity extraction

SKILLS

Cloud & Infrastructure: AWS (Sagemaker, S3 Bucket), Azure, GCP, Docker, MLOps, CI/CD Pipelines
Programming: Python (Pandas, NumPy, Scikit-learn), SQL, Pytorch, Tensorflow, REST API
Machine Learning & AI: Machine Learning, Deep Learning (RNN, CNN, LSTM), MLflow, NLP

PROJECTS

Volatility Forecasting with News Sentiment (LSE Capstone) — Python, Scikit-learn, Pandas, Regression

- An industry partner needed a robust 21-day-forward stock volatility forecast; architected a leakage-free ML pipeline using 10 years of price data, macro indicators (VIX) and LLM-scored news sentiment across 8 equities
- Engineered temporal features (rolling z-scores, holiday adjustments) and built/evaluated 5 model architectures, including Adaptive Lasso, Linear Regression, Optuna-tuned LightGBM, and an econometric GJR-GARCH baseline
- Validated results with moving-block bootstraps and feature-ablation testing against a demanding persistence baseline, reducing out-of-sample forecast RMSE by 13% (to 0.169)